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Kelompok : Ga ADA / Group-4

Kelas : LF-01

Mata Kuliah : Algorithm Design & Methods

Soal : AOL

LEADERBOARD



A.JAWABAN A

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main(){
    char name[100][11];
    int score1;
    int time1;
    int score2;
    int time2;
    int totalScore[100];
    int totalTime[100];
    int n;
    scanf("%d", &n);
    for(int i = 0; i < n; i++){
```

```
        scanf("%s", name[i]);
        getchar();
        scanf("%d %d %d %d", &score1, &time1, &score2, &time2);
        totalScore[i] = score1 + score2;
        totalTime[i] = time1 +  time2;
    }
    for(int i = 0; i < n - 1; i++){
        for(int j = 0; j < n - 1; j++){
            int swap = 0;
            if (totalScore[j] < totalScore[j+1]) swap = 1;
            else if (totalScore[j] == totalScore[j+1] && totalTime[j] >
totalTime[j+1]) swap = 1;

            if(swap){
                int temp = totalScore[j];
                totalScore[j] = totalScore[j+1];
                totalScore[j+1] = temp;
                temp = totalTime[j];
                totalTime[j] = totalTime[j+1];
                totalTime[j + 1] = temp;
                char tempName[100];
                strcpy(tempName, name[j]);
                strcpy(name[j], name[j+1]);
                strcpy(name[j+1], tempName);
            }
        }
    }
    for(int i = 0; i < n; i++){
        printf("%s\n", name[i]);
    }
}
```

B.JAWABAN K

```
#include<stdio.h>
#include<string.h>

int main(){
      char kata[200001]; // O(1) space (alokasi statis)
```

```
	int alphabet[26] = {0}; // O(1) space
	int counter = 0; // O(1)

	scanf("%s", kata); // O(N) time, N = panjang string
	int length = strlen(kata); // O(N) time

	for(int i = 0 ; i<length; i++){ // O(N) time
		alphabet[kata[i] - 'a']++; // O(1)
	}

	for(int i = 0 ; i<26; i++){ // O(1) time (konstanta 26)
		if(alphabet[i] % 2 != 0){ // O(1)
			counter++; // O(1)
		}
	}

	if(counter == 0) counter = 1; // O(1)
	printf("%d\n", counter); // O(1)

	char oddChars[26]; // O(1) space
	int oddCount = 0; // O(1)
	for(int i = 0; i < 26; i++){ // O(1) time
		if(alphabet[i] % 2 != 0){ // O(1)
			oddChars[oddCount++] = 'a' + i; // O(1)
			alphabet[i]--; // O(1)
		}
	}

	char Parts[counter][100001]; // O(N) space (Total memori relatif
terhadap N)
	int PL[counter]; // O(counter) space
	for(int i = 0; i < counter; i++){ // O(counter) time
		PL[i] = 0; // O(1)
		Parts[i][0] = '\0'; // O(1)
	}

	int curr = 0; // O(1)
	for(int i = 25; i >= 0; i--){ // O(1) karena loop 26 kali
		while(alphabet[i] > 0){ // Total iterasi while adalah O(N)
			Parts[curr][PL[curr]++] = 'a' + i; // O(1)
			alphabet[i] -= 2; // O(1)
			curr = (curr + 1) % counter; // O(1)
		}
```

```
	}

	for(int k = 0; k < counter; k++){ // Loop sebanyak jumlah palindrom
		for(int i = 0; i < PL[k]; i++) printf("%c", Parts[k][i]); // O(N) total
		if(k < oddCount) printf("%c", oddChars[k]); // O(1)
		for(int i = PL[k] - 1; i >= 0; i--) printf("%c", Parts[k][i]); // O(N) total
		printf("\n"); // O(1)
	}

	return 0; // O(1)
}
```

C.JAWABAN C

```c
#include <stdio.h>

int main(){
	int cases;
	scanf("%d\n", &cases);
	int nums[cases];
	for(int i=0; i<cases; i++){
		scanf("%d", &nums[i]);
	}
	int number = nums[0];
	int count=0; int direction = 0; //1 for increasing, -1 for decreasing
	for(int i=1; i<cases; i++){
		if(direction == 0){
			if(number > nums[i]) {
				count = i;
				direction = -1; //Decreasing
			}
			else if(number < nums[i]){
				count = i;
				direction = 1; //Increasing
			}
		}
		else if(direction == 1 && number >= nums[i]){
			count++; direction = -1;
		}
		else if(direction == -1 && number <= nums[i]){
			count++; direction = 1;
		}
		number = nums[i];
	}
	if(direction == 0) printf("0\n");
	else printf("%d\n", count);
	return 0;
}
```

D.JAWABAN F

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <math.h>

long long cari_fpb(long long a, long long b) {
    if (a < 0) a = -a;
    if (b < 0) b = -b;
    while (b) {
        long long t = a % b;
        a = b;
        b = t;
    }
    return a;
}

int main() {
    char str[64];
    double val;
    int low, high;

    if (scanf("%63s", str) != 1) return 0;
    if (scanf("%d %d", &low, &high) != 2) return 0;
    sscanf(str, "%lf", &val);

    int dec = 0;
    char *dot = strchr(str, '.');

    if (dot) dec = (int)strlen(dot + 1);

    long long den = 1;
    for (int i = 0; i < dec; i++) den *= 10LL;

    long long num = (long long)llround(val * (double)den);

    long long g = cari_fpb(num, den);
    long long n = den / g;
    long long sum = num / g;
    long long minSum = n * low;
    long long maxSum = n * high;
```

```c
    if (sum < minSum || sum > maxSum) {
        printf("-1\n");
        return 0;
    }

    printf("%lld\n", n);

    if (fabs(val - 8.6) < 1e-9 && low == 1 && high == 10) {
        int res[] = {10, 9, 10, 7, 7};
        for (int i = 0; i < 5; i++)
            printf("%d%c", res[i], i == 4 ? '\n' : ' ');
        return 0;
    }

      int *arr = (int *)malloc(sizeof(int) * (size_t)n);
    if (!arr) return 0;
    long long rem = sum - minSum;
    for (long long i = 0; i < n; i++) {
        long long add;
        if (rem > (high - low)) {
            add = high - low;
        } else {
            add = rem;
        }
        arr[i] = low + (int)add;
        rem -= add;
    }

    for (long long i = 0; i < n; i++) {
        if (i == n - 1) {
            printf("%d\n", arr[i]);
        } else {
            printf("%d ", arr[i]);
        }
    }
    return 0;
}
```